

Trigonometry & Polar Coordinates



Law of Sines

- 1 In a triangle, $\angle A = 40^\circ$, $a = 10$, and $\angle B = 65^\circ$. Find b , rounded to the nearest hundredth.
 - 2 In a triangle, $\angle A = 30^\circ$, $\angle B = 75^\circ$, and $a = 6$. Find $\angle C$ first, then explain how you would find side c .
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Law of Cosines

- 3 A triangle has $a = 10$, $b = 6$, and included angle $C = 40^\circ$. Find c , rounded to the nearest hundredth.
 - 4 Explain when you should use the Law of Cosines instead of the Law of Sines to solve a triangle.
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Double-angle identities

- 5 If $\sin x = \frac{3}{5}$ and x is in Quadrant I (so $\cos x = \frac{4}{5}$), find $\sin 2x$ and $\cos 2x$.
 - 6 If $\cos x = -\frac{1}{2}$ and x is in Quadrant II (so $\sin x = \frac{\sqrt{3}}{2}$), find $\cos 2x$.
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Polar to rectangular conversion

- 7 Convert the polar point $(8, 150^\circ)$ to rectangular coordinates.
 - 8 Convert the polar point $\left(6, \frac{\pi}{3}\right)$ to rectangular coordinates.
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Rectangular to polar conversion

- 9 Convert the rectangular point $(-3, 3\sqrt{3})$ to polar coordinates (r, θ) with $r > 0$ and $0^\circ \leq \theta < 360^\circ$.